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Suppression of crosstalk in fiber-optic gyroscopes with eight-state modulation method

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Abstract

Electrical crosstalk is one of the key factors affecting the accuracy of interferometric fiber-optic gyroscopes (FOGs), and conventional modulation methods have limitations in crosstalk suppression. This paper proposes a novel eight-state modulation (ESM) method aimed at effectively suppressing electrical crosstalk by eliminating the correlation between the modulation voltage and the demodulation sequence, thereby enhancing the gyroscope measurement accuracy. Compared with existing modulation techniques, the advantage of ESM lies in the fact that its crosstalk is independent of the step wave height. Theoretical analysis demonstrates that the ESM scheme can completely eliminate crosstalk with fixed phase delay at $\pi/2$ modulation depth. The experimental results validate the theoretical predictions: when operating at $\pi/2$ modulation depth, the FOG achieved a bias of approximately $0.030^\circ/\text{h}$, confirming the method's effectiveness.

1. Introduction

Interferometric fiber-optic gyroscopes (IFOGs) are extensively utilized as angular velocity sensors in diverse military and civilian domains [1–4]. In recent years, many research institutions have undertaken studies and designs focusing on integrated IFOGs (I-IFOGs) to meet the demand for miniaturization [5–7]. In general, IFOGs adopt a closed-loop configuration because of the significantly improved dynamic range and scale factor linearity [8]. In contrast to the open-loop detection system, the closed-loop FOG incorporates negative feedback to the open-loop system. The generated feedback phase difference compensates for the Sagnac phase shift and accomplishes closed-loop feedback control [8–10]. However, in closed-loop modulation, the rise of modulation crosstalk between the modulation signal and the detection loop causes the gyroscope to be insensitive to the rotation signal when measuring at low rotational speeds. In particular, for I-IFOG, a highly complex electromagnetic environment is introduced within circuits due to the miniaturization of circuitry, which leads to significant crosstalk and degradation of the I-IFOG performance. Traditional FOGs primarily employ two schemes: square-wave modulation (SWM) and four-state modulation (FSM). SWM has been widely adopted for its simplicity [4]. However, SWM suffers from significant electrical crosstalk between the modulation and demodulation circuits, particularly in low-speed or stationary applications where the Sagnac signal is extremely weak. To address the electrical crosstalk problem, FSM was developed [11, 12]. FSM eliminates modulation crosstalk during the demodulation process by generating statistically independent modulation and demodulation signals, avoiding the over-modulation crosstalk error introduced by the instantaneous switching. However, FSM is only applicable to the $\pi/2$ SWM phase. For high-precision FOGs, to suppress shot noise and relative intensity noise, over-modulation technology—characterized by a modulation phase ranging from $\pi/2$ to π (exceeding $\pi/2$)—is commonly employed to enhance the signal-to-noise ratio of the gyroscope's output signal [13–15].

In 2018, Zhang *et al* from Beihang University, China, first proposed the six-state modulation (SSM) technology. This innovative scheme achieves the separation of the fundamental frequency of the modulation voltage and the demodulation reference frequency by increasing the number of states per modulation cycle from four to six. The proposed scheme theoretically eliminates the electrical crosstalk completely at a modulation depth of $2\pi/3$, achieving a zero-dead-zone result in experiments [16]. In 2024, a research team from Beijing University of Posts and Telecommunications further advanced the SSM technology and proposed an improved scheme based on flexible modulation voltage parameters. This new solution theoretically achieves a reduction of angular random walk (ARW) over 60%, while the zero-bias instability reaches $0.0035^\circ/\text{h}$ [17].

The analysis above reveals that multi-state schemes cannot simultaneously achieve : (1) zero-crosstalk operation at the standard modulation depth of $\varphi_m = \pi/2$, (2) analytical sequence design with mathematically guaranteed zero-point for predictable performance, and (3) practical implementation with moderate complexity. This gap motivates our work. Based on the current technological status, this paper proposes an eight-state modulation (ESM) scheme. This solution sets the zero-interference operating point at the $\pi/2$ modulation depth, effectively addressing the engineering compatibility issues of existing technologies. Simultaneously, it fundamentally suppresses output oscillations caused by system parameter variations. Experimental verification demonstrates that the ESM achieves an outstanding bias stability of approximately $0.030^\circ/\text{h}$ at $\pi/2$ modulation depth, providing a new technical pathway for the further development and industrial application of FOG modulation technology.

2. Principle and analysis of digital closed-loop control system

2.1. Principle of I-IFOG

The I-IFOG consists of a super-luminescent diode (SLD) light source, a coupler, a Ge-on-Si photodetector, a phase modulator, and a fiber coil. When the IFOG is fully integrated and operational, the micro light source drive circuit activates the on-chip SLD light source. Initially, the light is split into two beams of equal intensity. Subsequently, these two beams pass through a LiNbO_3 waveguide for polarization, then undergo push-pull phase modulation. The signal processing circuit applies a periodic ramp modulation signal to the phase modulator. The processing circuit serves two purposes. First, it introduces a periodic phase offset to the light signal, thereby adjusting the operating point of the gyroscope. Second, it applies a step waveform to the phase modulator to achieve closed-loop compensation for the angular velocity phase shift. Following this, the two beams propagate along the clockwise and counter-clockwise directions through the micro polarization-maintaining fiber ring.

Due to the Sagnac effect, a phase difference $\Delta\varphi_s$ proportional to the rotational angular velocity Ω is generated between a pair of counter-propagating beams. The phase difference between the two beams for one trip $\Delta\varphi_s$ is calculated by [4]

$$\Delta\varphi_s = \frac{2\pi LD}{\lambda_c} \Omega \quad (1)$$

where D is the coil equivalent diameter, L is the coil length, and λ_c is the average source wavelength.

After exiting the fiber ring, the two beams interact with each other through interference at the Y-branch, and the interference intensity is detected by the on-chip detector. Finally, the micro signal processes the circuit samples and the light intensity signal to demodulate the angular velocity.

Due to the considerable distance between circuit components in traditional IFOGs, the electronic cross-coupling is extremely weak. The gyroscope error caused by circuit crosstalk is negligible. However, the miniaturization of signal processing circuits results in a very small space between circuit components. The electromagnetic environment within the gyroscope's integrated circuit and across interconnected circuit boards grows remarkably intricate, exacerbating signal crosstalk to non-negligible levels.

2.2. Mechanism analysis of phase shift error caused by circuit crosstalk

The primary circuit crosstalk in IFOGs is attributed to the modulation circuitry [18, 19]. As illustrated in figure 1, the signal processing circuit employs a digital-to-analog conversion process to generate a modulation voltage for phase modulation. In the practical operation of the I-IFOG, the Y-waveguide modulator is driven by a digitally synthesized ramp voltage, which exhibits precise linear proportionality to the half-wave voltage, while the demodulation signal is a weak voltage transformed from the interference intensity detected by the photodetector. Critically, the modulation and demodulation frequencies are intentionally matched to maximize signal recovery, while a significant electrical crosstalk is introduced through electromagnetic field interactions in compact layouts.

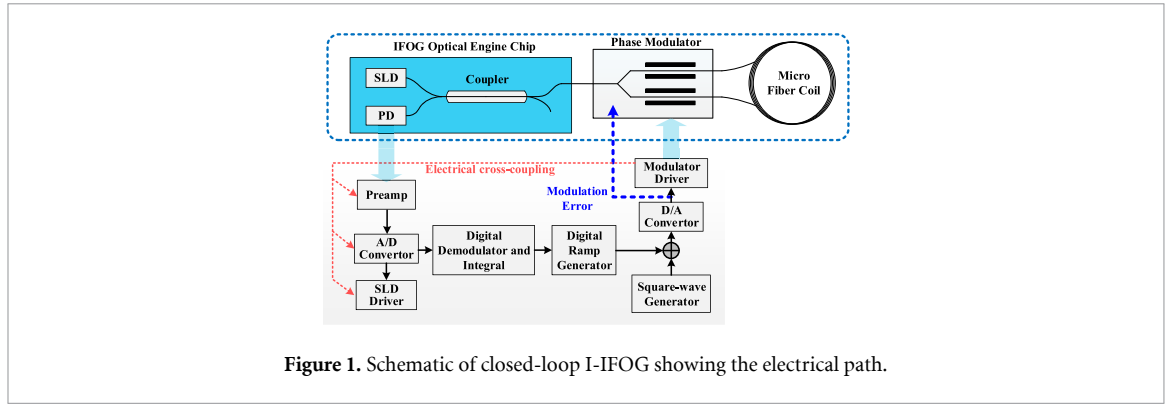


Figure 1. Schematic of closed-loop I-IFOG showing the electrical path.

Figure 1 presents the crosstalk paths in the integrated gyroscope circuit section. A portion of the modulation signal couples into the photodetector output via electrical cross-coupling mechanisms, introducing additional interference terms. Following A/D conversion, the interference signal contains both rotational speed information and induced circuit crosstalk components. A non-reciprocity phase shift error arises after demodulation. The demodulation-induced modulation error propagates to the phase modulator through signal coupling, consequently generating dead-zone effects in the system response [20, 21]. This investigation employs a closed-loop feedback control architecture utilizing conventional SWM to formulate a mathematical model characterizing phase shift errors induced by crosstalk interference in IFOG circuits. The crosstalk-induced phase shift (denoted as $\varphi_{cr}(t)$) exhibits dependencies on the amplitude spectrum of the modulation voltage and the distributed coupling coefficient of the crosstalk pathway [8].

We consider a bidirectional SWM with discrete modulation depths $\{\varphi_m, -\varphi_m\}$, where the modulation depth φ_m is set at $\pi/2$ radians. The responding demodulation sequence is $\{1, -1\}$. Each state lasts for τ (τ is the transition time of the fiber ring). During the temporal interval corresponding to the modulation depth of φ_m , the photodetector's output intensity $I'(t)$ is expressed as

$$I'(t) = I(t) \{1 + \cos[\Delta\varphi_s + \varphi_m + \varphi_f + \varphi_{cr}(t)]\}. \quad (2)$$

The compensating phase shift φ_f , which arises from the modulated step wave, is extracted through demodulation of the interference across successive modulation periods, enabling real-time compensation for the Sagnac phase shift. The demodulated light intensity is expressed as

$$\begin{aligned} \Delta I = I(t) \{1 + \cos[\Delta\varphi_s + \varphi_m + \varphi_f + \varphi_{cr}(t)]\} \\ - I(t + \tau) \{1 + \cos[\Delta\varphi_s - \varphi_m + \varphi_f + \varphi_{cr}(t + \tau)]\}. \end{aligned} \quad (3)$$

The implemented closed-loop operation presupposes stationary conditions, where (1) the FOG experiences constant rotation input, and (2) the optical source exhibits intensity stationarity during a single demodulation period ($I(t + \tau) = I(t) = I_0$). Based on the closed-loop feedback conditions ($\Delta I = 0$ when stable), the feedback phase shift can be obtained as

$$\varphi_f = -\Delta\varphi_s - \frac{\varphi_{cr}(t) + \varphi_{cr}(t + \tau)}{2}. \quad (4)$$

The non-reciprocity phase shift error caused by electrical crosstalk can be mathematically expressed as

$$\varphi_e = -\varphi_f - \Delta\varphi_s = \frac{\varphi_{cr}(t) + \varphi_{cr}(t + \tau)}{2}. \quad (5)$$

The bias modulation is generated by the differential phase shift $\Delta\Phi_m(t)$ between the modulation phase $\Phi_m(t)$ and the modulation phase delayed by time by τ , $\Phi_m(t - \tau)$, which is expressed as

$$\Delta\Phi_m(t) = \Phi_m(t) - \Phi_m(t - \tau). \quad (6)$$

The relationship between the I-IFOG modulation voltage $V_m(t)$ and the modulation phase $\Phi_m(t)$ is

$$V_m(t) = K_m(t) \Phi_m(t) \quad (7)$$

Table 1. The four-state modulation sequence and double closed-loop demodulation sequence.

State	Modulation phase shift $\Delta\Phi_m(t)$	Total modulated phase	$S_1(n)$	$S_2(n)$
D1	φ_m	$\varphi_m + \varphi_{stp}$	-1	-1
D2	$2\pi - \varphi_m$	$2\pi - \varphi_m + \varphi_{stp}$	1	1
D3	$-\varphi_m$	$\varphi_m + \varphi_{stp} + \varphi_f$	1	-1
D4	$\varphi_m - 2\pi$	$\varphi_m - 2\pi + \varphi_{stp} + \varphi_f$	-1	1

$S_1(n)$ and $S_2(n)$ are the demodulation sequences of the first and second loops, respectively.

where $K_m(t)$ is the modulation gain of the integrated phase modulator. The crosstalk voltage exhibits a linear dependence on the modulation voltage, with negligible phase delay effects establishing the relationship as

$$V_e(t) = K_c V_m(t) + V_{e0} \quad (8)$$

where K_c is the crosstalk coupling coefficient and V_{e0} is a constant term that does not affect the final output of the gyroscope after demodulation. The crosstalk voltage is expressed as

$$V_e(t) = K_c K_m(t) \Phi_m(t). \quad (9)$$

The manifestation of crosstalk-induced non-reciprocal phase errors in I-IFOG systems is determined by the correlation properties between the crosstalk voltage and the demodulation sequence. A quantitative assessment of crosstalk effects can be obtained by calculating the cross-correlation function between these two signals:

$$\gamma = \frac{1}{N} \sum_{n=0}^N V_e(n) S(n-m) \quad (10)$$

where $S(n)$ is the demodulation sequence and m is the phase delay between the crosstalk voltage and the demodulation sequence. Substituting formulas (8) and (9) into formula (10), the cross-correlation function is mathematically expressed as

$$\gamma = \frac{1}{N} K_m K_c \sum_{n=0}^N \Phi_m(n) S(n). \quad (11)$$

Through cross-correlation analysis between the demodulation sequence and crosstalk interference in a closed-loop configuration, the system-level impact of circuit crosstalk can be quantitatively characterized.

2.3. FSM

The FSM method introduces two additional modulation phases based on SWM with the same modulation period. This approach optimizes the second closed-loop feedback circuit while preserving the demodulation performance of the gyroscope rotation rate output in the first closed-loop system. The modulation sequences of the four-state SWM method commonly used in double closed-loop IFOG and the demodulation sequences are shown in table 1.

The closed-loop IFOG employs a dual-output phase modulator in a push-pull configuration to implement its modulation scheme, where φ_m represents the modulation depth of each state. φ_{stp} is the initial step phase of the digital step wave superimposed on the modulated waveform, while φ_f represents the height of the steps corresponding to the closed-loop feedback compensation. A comparison of figures 2(a) and (b) reveals distinct demodulation references between the primary and secondary servo loops. The misalignment signals are generated per half-period τ in the FSM scheme, which is different from the traditional two square-wave modulation's rate-sensitive error detection. These misalignment signals can be detected with the demodulation sequence at $2f_p$ ($f_p = 1/(2\tau)$). Critically, the secondary loop operates at a fixed high frequency, decoupled from the primary loop.

Through the configuration, the resulting phase modulation sequence at the gyroscope output can be mathematically described as

$$\Phi_m(n) = \left\{ \dots, \frac{\varphi_m}{2} + \varphi_{stp}, \pi - \frac{\varphi_m}{2} + \varphi_{stp}, -\frac{\varphi_m}{2} + \varphi_{stp} + \varphi_f, \frac{\varphi_m}{2} - \pi + \varphi_{stp} + \varphi_f, \dots \right\}. \quad (12)$$

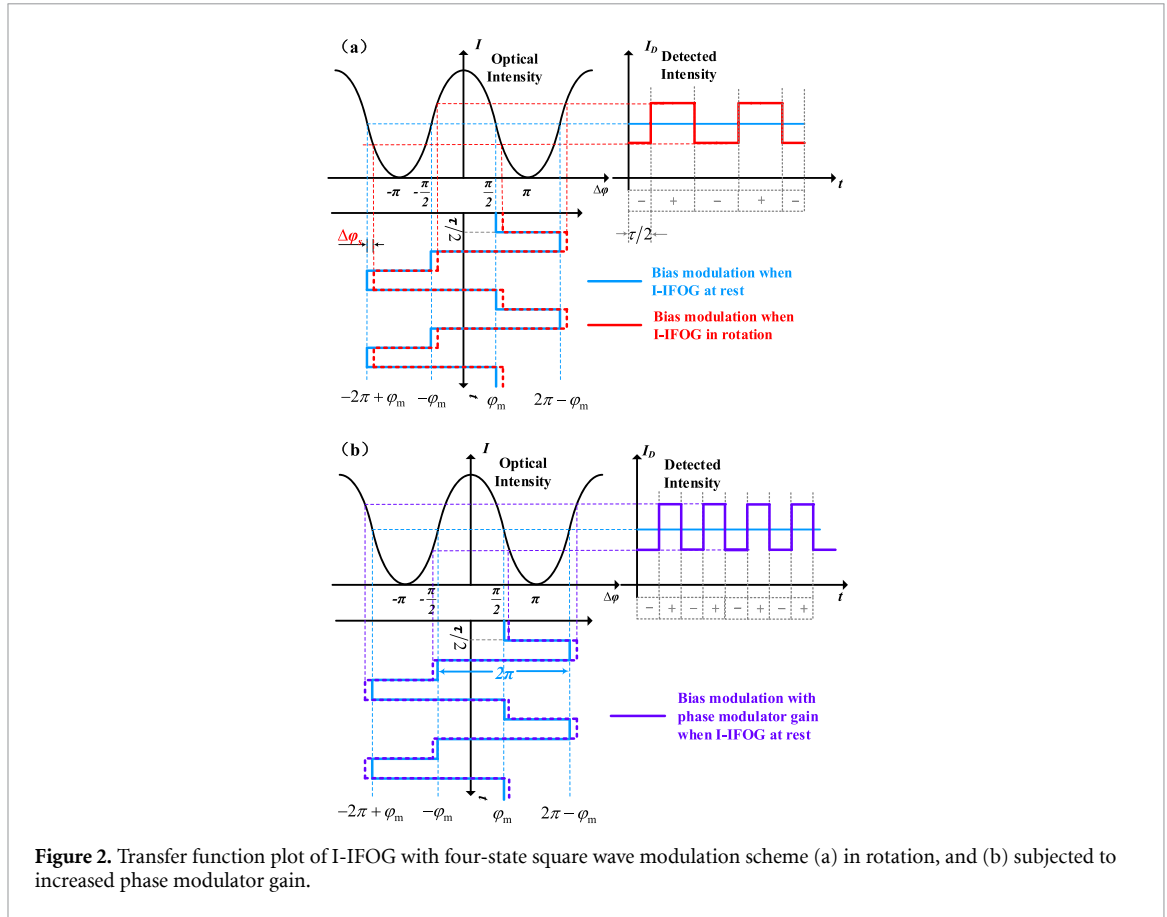


Figure 2. Transfer function plot of I-IFOG with four-state square wave modulation scheme (a) in rotation, and (b) subjected to increased phase modulator gain.

Given the detected interference intensities $I(t_{n+i})$ ($i = 0, 1, 2, 3$) corresponding to the four modulation states at the photodetector, the demodulated outputs for the first and second closed-loop feedback paths are expressed as

$$\begin{cases} \Delta I_1 = -I(t_n) + I(t_{n+1}) + I(t_{n+2}) - I(t_{n+3}) \\ \Delta I_2 = -I(t_n) + I(t_{n+1}) - I(t_{n+2}) + I(t_{n+3}) \end{cases} \quad (13)$$

The respective output sequences of $S_1(n)$ and $S_2(n)$ corresponding to the primary and secondary feedback paths are expressed as

$$\begin{aligned} S_1(n) &= \{\dots, -1, 1, 1, -1, \dots\} \\ S_2(n) &= \{\dots, -1, 1, -1, 1, \dots\} \end{aligned} \quad (14)$$

When a correlation between the demodulation sequence and modulation sequence exists, cross-interference will occur in the circuit. According to formula (11), the cross-correlation function of the dual closed loop is expressed as

$$\begin{cases} \gamma_1 = \frac{1}{N} K_m K_c \sum_{n=0}^N \Phi_m(n) S_1(n) \\ \gamma_2 = \frac{1}{N} K_m K_c \sum_{n=0}^N \Phi_m(n) S_2(n) \end{cases} \quad (15)$$

where K_m is the modulation gain of the phase modulator and K_c is the crosstalk coupling efficient.

Table 2. The dual-loop eight-state modulation sequence.

State	Modulation phase shift $\Delta\Phi_m(t)$	Total modulated phase	$S_1(n)$	$S_2(n)$
D1	φ_m	$\varphi_m/2 + \varphi_{stp}$	-1	0
D2	$2\pi - \varphi_m$	$\pi - \varphi_m/2 + \varphi_{stp}$	1	1
D3	$-\varphi_m$	$-\varphi_m/2 + \varphi_{stp}$	1	0
D4	φ_m	$\varphi_m/2 + \varphi_{stp}$	-1	-1
D5	$-\varphi_m$	$-\varphi_m/2 + \varphi_{stp} + \varphi_f$	1	0
D6	$\varphi_m - 2\pi$	$\varphi_m/2 - \pi + \varphi_{stp} + \varphi_f$	-1	1
D7	φ_m	$\varphi_m/2 + \varphi_{stp} + \varphi_f$	-1	0
D8	$-\varphi_m$	$-\varphi_m/2 + \varphi_{stp} + \varphi_f$	1	-1

$S_1(n)$ and $S_2(n)$ are the demodulation sequences of the first and second loops, respectively.

Substituting equations (12) and (14) into (15), the cross-correlation functions of the phase modulation sequence and demodulation sequence are calculated as

$$\begin{cases} \gamma_1 = \frac{1}{2}K_m K_c (\pi - \phi_m) \\ \gamma_2 = 0 \end{cases} \quad (16)$$

Equation (16) demonstrates that the cross-correlation function γ_2 equals zero, which confirms that the circuit crosstalk induced by the modulation signal has no impact on the gain error control of the second closed-loop system. Meanwhile, for the first closed-loop system, γ_1 is linearly related to the modulation depth φ_m . Theoretically, γ_1 equals zero when φ_m equals π where the gyro output sensitivity is zero, which is unreasonable for actual gyro application. Consequently, the demodulation performance of FSM systems is invariably compromised by cross-interference effects.

3. Suppression of crosstalk with ESM scheme

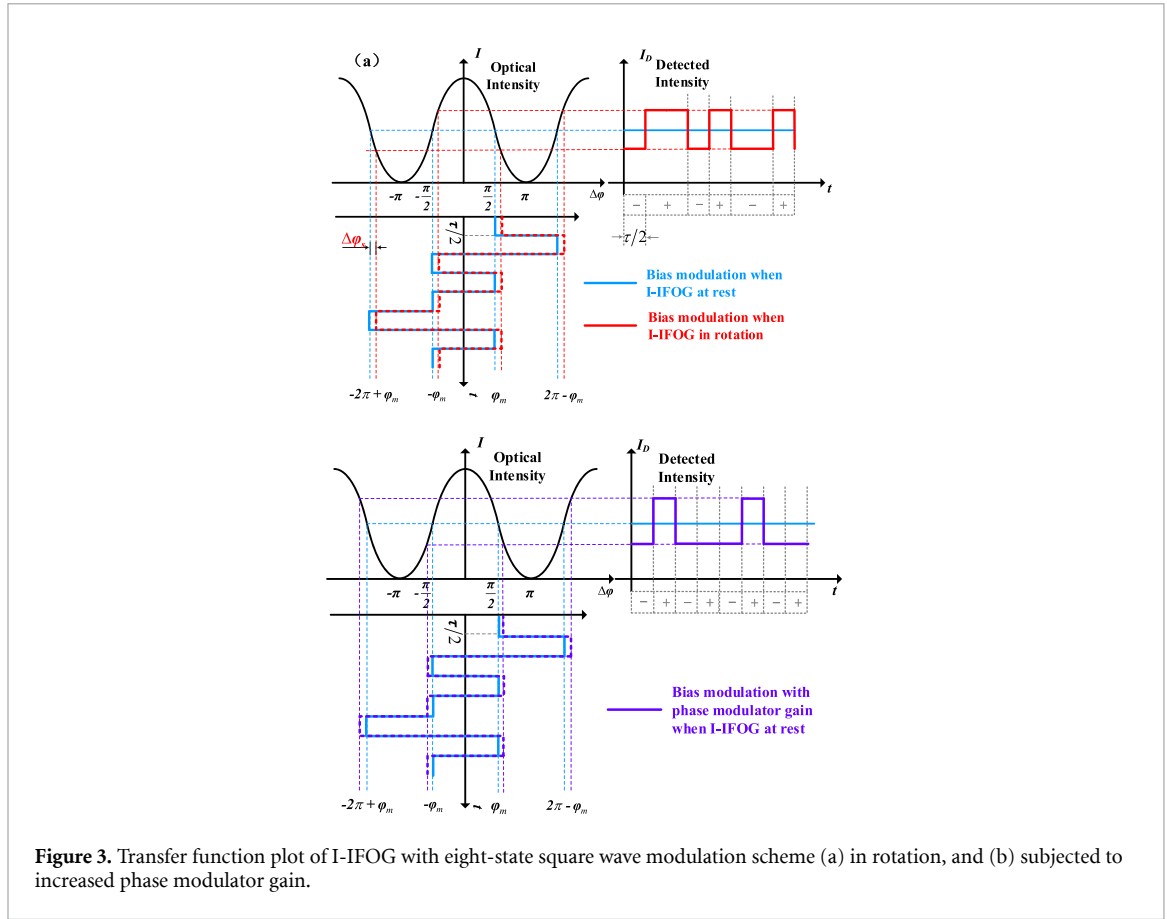
Finite state modulation is currently the most efficient suppression method. Optimized modulation sequences actively suppress spectral overlap between the output signal and demodulation reference, thereby lowering correlation-induced distortion. While conventional FSM mitigates the correlation between the modulation voltage and demodulation sequence through the increased modulation depth, a non-negligible correlation persists within the IFOG's optimal modulation regime, leaving residual circuit crosstalk uncompensated.

To minimize the effect of circuit crosstalk on the demodulation, we propose the ESM method for I-IFOG. The ESM method features a reduced number of states, offering simplicity and ease of implementation. It imposes minimal demands on the hardware performance and does not compromise the other performance aspects of the I-IFOG. Theoretically, at the optimal modulation depth, the correlation between the modulated signal and the demodulation sequence can be minimized to zero, thereby fully suppressing the non-reciprocity error induced by circuit crosstalk. The design methodology is based on FSM and SSM [16, 17], implemented by superimposing the eight-state sequence $(0, 2\pi - 2\varphi_m, -2\varphi_m, 0, 0, 2\varphi_m - 2\pi, 2\varphi_m, 0)$ and the corresponding square wave sequence $(\varphi_m, -\varphi_m)$, where each state of the eight-state sequence lasts for $\tau/4$. Note that the superimposed 2π phase shift does not change $\Delta\Phi_m(t)$, and the voltage fundamental frequency is separated from the demodulation sequence frequency. The detailed modulation sequence is presented in table 2.

The corresponding ESM sequence can be expressed as

$$\Phi_m(n) = \left\{ \begin{array}{l} \cdots, \frac{\varphi_m}{2} + \varphi_{stp}, \pi - \frac{\varphi_m}{2} + \varphi_{stp}, -\frac{\varphi_m}{2} + \varphi_{stp}, \\ \frac{\varphi_m}{2} + \varphi_{stp}, -\frac{\varphi_m}{2} + \varphi_{stp} + \varphi_f, \frac{\varphi_m}{2} - \pi + \varphi_{stp} + \varphi_f, \\ \frac{\varphi_m}{2} + \varphi_{stp} + \varphi_f, -\frac{\varphi_m}{2} + \varphi_{stp} + \varphi_f \cdots \end{array} \right\}. \quad (17)$$

The transfer functions of the interference light intensity with ESM are depicted in figure 3.



From the transfer function plot, the intensity of each state in the closed-loop modulation is obtained as

$$\begin{aligned}
 D_1 = D_4 = D_7 &= \frac{K_0}{2} [1 + \cos(\Delta\varphi_s + \varphi_1 + \varphi f)] \\
 &= \frac{K_0}{2} [1 + \cos \varphi_1 \cos(\Delta\varphi_s + \varphi f) - \sin \varphi_1 \sin(\Delta\varphi_s + \varphi f)] \\
 D_3 = D_5 = D_8 &= \frac{K_0}{2} [1 + \cos(\Delta\varphi_s - \varphi_1 + \varphi f)] \\
 &= \frac{K_0}{2} [1 + \cos \varphi_1 \cos(\Delta\varphi_s + \varphi f) + \sin \varphi_1 \sin(\Delta\varphi_s + \varphi f)] \\
 D_2 &= \frac{K_0}{2} [1 + \cos(\Delta\varphi_s + \varphi_2 + \varphi f)] \\
 &= \frac{K_0}{2} [1 + \cos \varphi_2 \cos(\Delta\varphi_s + \varphi f) - \sin \varphi_2 \sin(\Delta\varphi_s + \varphi f)] \\
 D_6 &= \frac{K_0}{2} [1 + \cos(\Delta\varphi_s - \varphi_2 + \varphi f)] \\
 &= \frac{K_0}{2} [1 + \cos \varphi_2 \cos(\Delta\varphi_s + \varphi f) + \sin \varphi_2 \sin(\Delta\varphi_s + \varphi f)].
 \end{aligned} \tag{18}$$

where D_i ($i = 1, 2, \dots, 8$) are corresponding to each state, K_0 is the forward channel gain, $\varphi_1 = \varphi_m$, $\varphi_2 = 2\pi - \varphi_m$. Substituting φ_1 and φ_2 into each state, the relationship between each state is expressed as

$$\begin{aligned}
 D_1 &= D_4 = D_6 = D_7 \\
 D_2 &= D_3 = D_5 = D_8
 \end{aligned} \tag{19}$$

The digital demodulation is obtained as

$$\begin{cases}
 D_{rate} = -D_1 + D_2 + D_3 - D_4 + D_5 - D_6 - D_7 + D_8 \\
 \quad = 4K_0 \sin \varphi_m \sin(\varphi_s + \varphi f) \\
 D_{gain} = D_2 - D_4 + D_6 - D_8
 \end{cases} \tag{20}$$

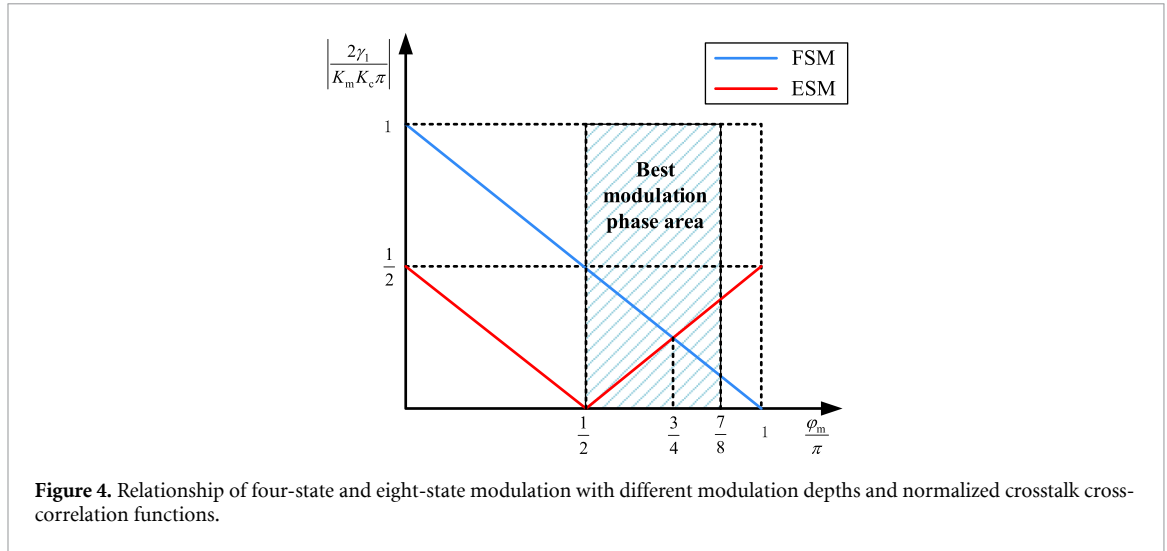


Figure 4. Relationship of four-state and eight-state modulation with different modulation depths and normalized crosstalk cross-correlation functions.

The phase modulator's gain error is detected and compensated during each modulation half-period τ . The modulation sequence of the first loop and the second loop is

$$\begin{cases} S1(n) = \{\dots, -1, 1, 1, -1, 1, -1, -1, 1, \dots\} \\ S2(n) = \{\dots, 0, 1, 0, -1, 0, 1, 0, -1, \dots\} \end{cases} \quad (21)$$

The two demodulation sequences are orthogonal to each other, which means that the second servo loop works independently of the main control loop. As derived in equation (15), the cross-correlation function between the ESM crosstalk sequence and the demodulation sequence in the zero-phase delay state can be obtained as

$$\begin{cases} \gamma_1 = \frac{1}{2}K_mK_c\left(\frac{\pi}{2} - \phi_m\right) \\ \gamma_2 = 0 \end{cases} \quad (22)$$

The curves corresponding to the different modulation depths and normalized crosstalk correlations for the traditional four-state and proposed ESM methods are plotted in figure 4. The crosstalk correlation function γ_1 of the first closed loop in ESM has a linear relationship with the modulation depth. The absolute value is symmetric about $\pi/2$ where $\gamma_1 = 0$, and the demodulation integrity in the first closed-loop circuit remains unaffected by circuit crosstalk. Theoretically, it is possible to completely suppress polarization crosstalk errors while ensuring the maximum sensitivity of the I-IFOG. For high-performance IFOGs, the choice of modulation depth is critical to the system performance. Theoretically, when the modulation depth is 0 or π , the detection sensitivity approaches zero, and the optimal modulation depth for maximum detection sensitivity is $\pi/2$ [4]. Considering the crosstalk correlation, the optimal modulation depth should range from $\pi/2$ to π . However, excessively low sensitivity would lead to a drastic increase in measurement errors for the IFOG. Therefore, the modulation depth should not approach π . As reported in the literature [15, 16], the optimal modulation depth for IFOGs typically lies in the range of $\pi/2$ to $7\pi/8$.

Furthermore, within the optimal modulation depth region, except for the range of $3\pi/4 \sim 7\pi/8$, the eight-state modulated signal exhibits lower correlation with crosstalk signals compared to traditional FSM, demonstrating superior polarization error suppression capability. The correlation function of the second closed loop γ_2 equals zero. Circuit crosstalk generated by the modulation will not affect the second closed-loop demodulation, meaning it will not impact the suppression of the modulation error. Moreover, the cross-correlation function of the dual closed-loop system is independent of both the initial step phase φ_{stp} and the step height φ_f of the modulation signal.

While ideal models indicate negligible phase delay between the crosstalk and modulated signals, practical implementation in IFOGs consistently reveals residual phase shifts. Figure 5 characterizes the normalized crosstalk error as a function of the crosstalk delay phase for four-state and eight-state architectures. Given the 360° periodicity and bilateral symmetry about 180° inherent to the cross-correlation function, the phase characterization was confined to the $[0^\circ, 180^\circ]$ interval. The figure demonstrates that

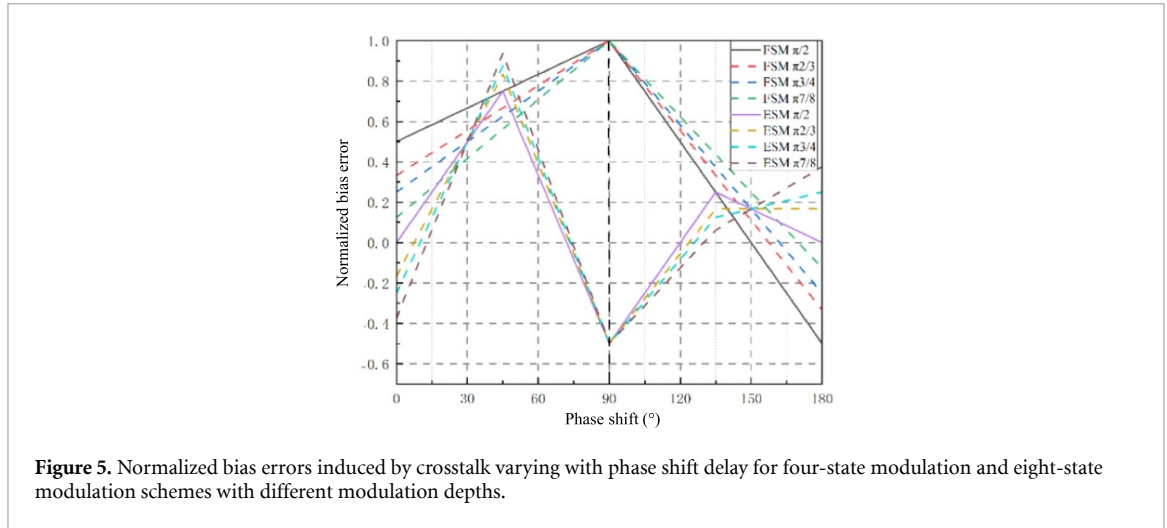


Figure 5. Normalized bias errors induced by crosstalk varying with phase shift delay for four-state modulation and eight-state modulation schemes with different modulation depths.

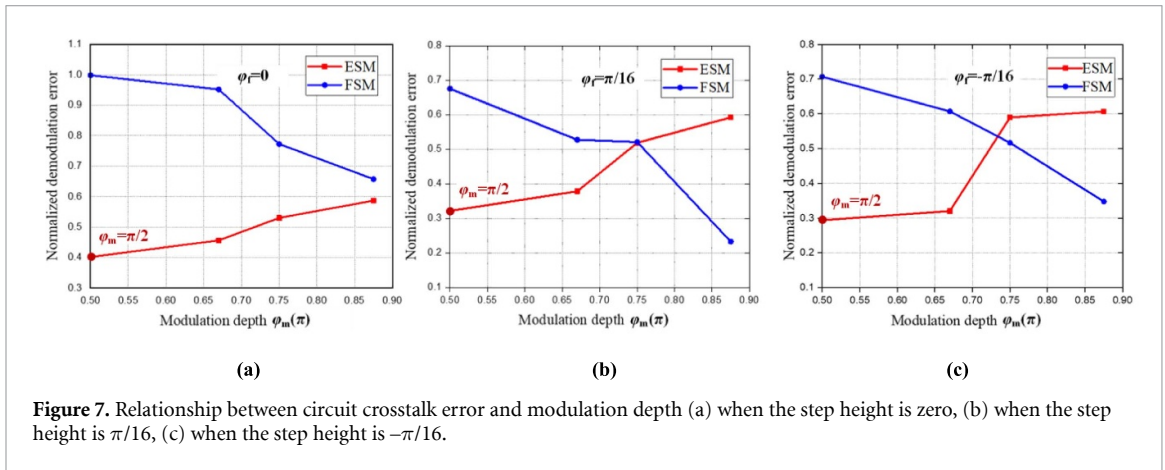
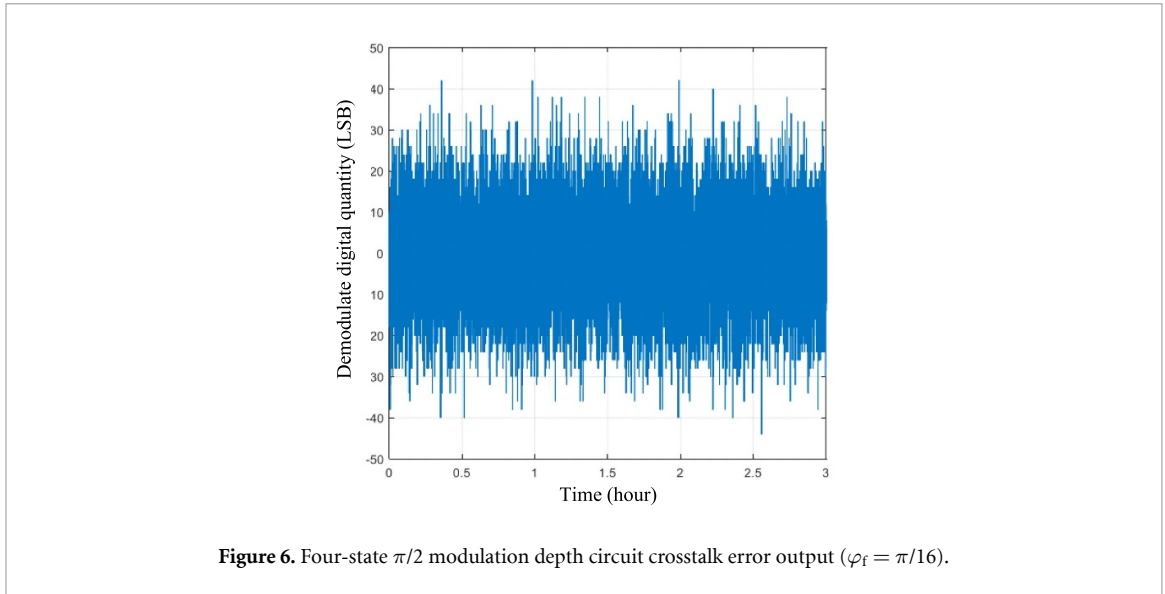
ESM achieves a near-zero crosstalk error ($\delta < 0.05\%$) within the phase intervals 0° – 10° and 120° – 130° , and consistently at precisely 75° —regardless of the modulation depth. Across the full delay phase spectrum (0° – 180°), ESM reduces the crosstalk error compared to FSM.

The proposed ESM scheme offers several notable advantages. Across the entire phase delay range, it demonstrates superior polarization error suppression capabilities, potentially achieving complete elimination of circuit crosstalk errors. In practical circuit operation, the crosstalk-induced voltage exhibits minimal phase delay relative to the modulation signal, typically approximating zero phase delay [22]. The ESM scheme with a $\pi/2$ modulation depth enables simultaneous optimization of effective circuit crosstalk suppression and maximized gyroscope sensitivity based on theoretical analysis. In addition, the cross-correlation function of the dual closed-loop system exhibits independence from both the initial phase φ_{stp} and step height φ_f of the modulation signal, which effectively suppresses periodic oscillations in the gyroscope output caused by circuit crosstalk variation during the periodic reset of the modulation ramp signal.

4. Experiment

To verify the effect of the ESM scheme on crosstalk suppression, a closed-loop I-IFOG was constructed based on figure 1. The experiment utilized a 1310 nm wavelength SLD light source, with the fiber coil wound using a 60/100 μm ultra-fine diameter polarization-maintaining fiber, totaling a length of 620 m. Our gyroscope was fixed on a marble surface at a latitude of 45.77° under ambient temperature without active control. The experiment commenced by setting the Sagnac phase step $\varphi_f = 0$. Under controlled conditions, both the four-state and ESM schemes were applied with the modulation depths φ_m sequentially configured to $\pi/2$, $2\pi/3$, $3\pi/4$, and $7\pi/8$. For each parameter set, the primary closed-loop demodulator output (characterizing signal crosstalk error) was recorded.

Figure 6 presents the circuit crosstalk error output data for the traditional FSM when the step wave height $\varphi_f = 0$ and the modulation depth $\varphi_m = \pi/2$. We configured the I-IFOG to operate in open-loop mode, where the demodulated output of the first loop circuit is directly used as the gyroscope output. The gain control output of the second loop circuit was set to a fixed ideal value calibrated during closed-loop mode operation. When both the FSM and ESM schemes are employed, the demodulated output of the I-IFOG should theoretically be zero. In practice, however, due to circuit crosstalk errors, the gyroscope exhibits a non-zero demodulated output. The voltage signal converted from the photodetector output, along with the demodulated output of the I-IFOG, was recorded via a serial port. By averaging 10 800 demodulation cycles and taking the absolute value of the result, the relationship between the circuit crosstalk error and modulation depth was derived, as shown in figure 7. From the figure, it can be seen that with FSM, as the modulation depth increases, the circuit crosstalk error gradually decreases, while the crosstalk error gradually increases with ESM. Across the evaluated modulation



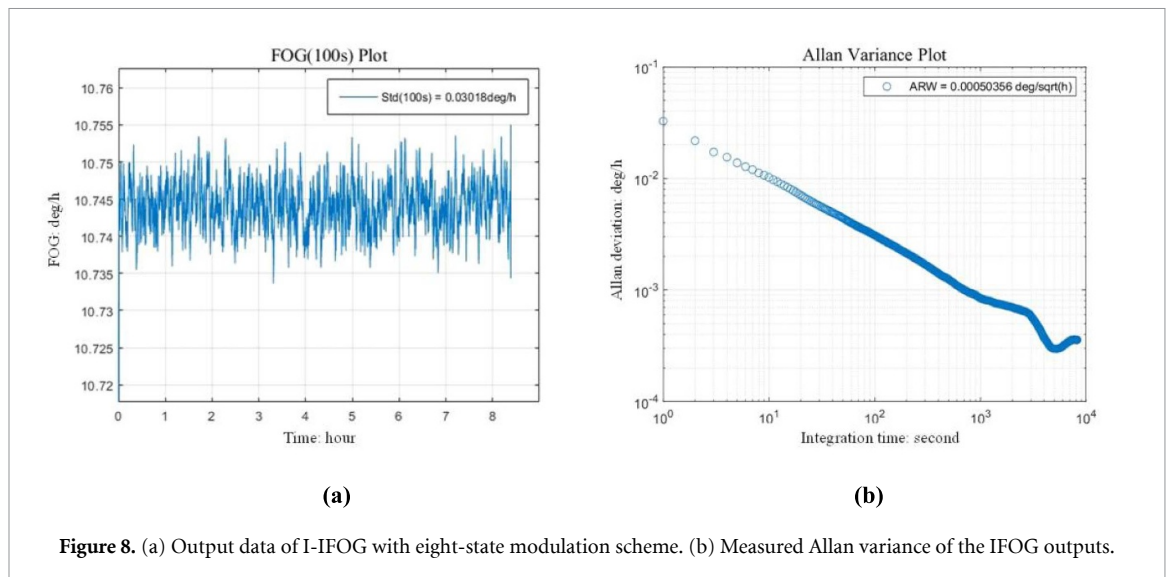
depths, ESM demonstrates a superior signal integrity, exhibiting a lower crosstalk error than the four-state schemes. When $\varphi_f = 0$, the ESM with a modulation depth of $\pi/2$ exhibits the minimum circuit crosstalk error, as shown in figure 7(a).

To systematically probe the correlation between modulation schemes and feedback phase step amplitude, we implemented discrete phase offsets of $\varphi_f = \pm \pi/16$ rad as controlled variables. The circuit crosstalk error versus the modulation depth when the step wave height is non-zero is depicted in figures 7(b) and (c). Within the modulation depth range of $\pi/2$ to $3\pi/4$, the circuit crosstalk error of ESM is smaller than that of FSM. Moreover, ESM achieves a minimal circuit crosstalk error at $\pi/2$ modulation depth and maintains error invariance to height (φ_f) variations in the feedback staircase waveform.

In summary, the experimental tests show that the phase delay between the circuit crosstalk signal and the modulation signal is minimal in the implemented I-IFOG. Employing the proposed ESM scheme, this study demonstrates an effective reduction of the correlation between the modulated signal and demodulation sequence for a modulation depth of $\pi/2$.

The I-IFOG prototype is fabricated with an EMC-optimized circuit board. The static performance characterization of 620 m I-IFOG including angular random walk (ARW) and bias stability is tested via earth rotation rate measurement gyro output data using the proposed technique. The tested data are averaged over 100 s to reduce noise. Figure 8 demonstrates the gyro output data and measured Allan variance plot.

As shown in figure 8, the bias stability is determined to be $0.03018^\circ/\text{h}$ with a 100 s smoothing interval, and the ARW is $0.005^\circ/\sqrt{\text{h}}$.



5. Conclusion

This paper proposes a novel ESM method for application in I-IFOGs. The proposed modulation scheme suppresses electrical crosstalk by eliminating the correlation between the modulation voltage and the demodulation sequence, thereby enhancing the gyroscope accuracy. A key advantage of this ESM is that its crosstalk error is independent of the step wave height, effectively preventing periodic oscillations in the gyroscope output caused by circuit crosstalk. Theoretically, the ESM scheme can completely eliminate crosstalk, exhibiting a fixed phase delay at $\pi/2$ modulation depth. The experimental results confirm this: when operating at a modulation depth of $\pi/2$, the gyroscope employing the ESM scheme achieved a bias stability of approximately 0.030°/h. In conclusion, the proposed ESM strategy theoretically enhances the gyro signal demodulation accuracy and suppresses noise, but the high complexity of ESM and demodulation conflicts with the miniaturization of FOG microsystems. To address this limitation, future research will focus on optimizing the modulation and demodulation architecture to fulfill the practical application requirements of modulation–demodulation modules for miniaturized and high-precision FOGs.

Data availability statement

The data cannot be made publicly available upon publication because they are not available in a format that is sufficiently accessible or reusable by other researchers. The data that support the findings of this study are available upon reasonable request from the authors.

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